Amrita School of Engineering

Bengaluru Campus

Department of Electronics & Communication Engineering

23ECE387 Embedded Systems Laboratory

23ECE313 Embedded Systems

VI Semester ECE

System Level Report

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AIM:

To design and implement an ARM Cortex-M4 based smart staircase lighting automation system using dual IR motion direction sensors to detect human movement and direction, and to automatically control staircase lights through relay switching for energy-efficient operation.

Components Required:

Name	Quantity
STM32F446RE	1
IR Sensors	2
Relay Module	1
LEDs	4
Resistors and Wire	

Theory:

The proposed system implements a smart staircase lighting automation technique using an ARM Cortex-M4 based controller, dual IR motion sensors, LEDs, and a relay module. The main objective of the system is to automatically control staircase lights based on the direction of human movement, thereby improving convenience and reducing unnecessary power consumption.

Two IR motion sensors, namely S1 and S2, are placed at opposite ends of the staircase. These sensors detect the presence and movement direction of a person. When a person approaches from the bottom side of the staircase, sensor S1 gets triggered first. The controller interprets this as upward movement and sequentially turns ON the staircase lights from LED1 to LED5, creating a progressive lighting effect in the upward direction. Similarly, when a person approaches from the top side, sensor S2 gets activated first. The controller then identifies the movement as downward direction and turns ON the lights in reverse order, from LED5 to LED1

LEDs stand for staircase lights, whereas the fifth light is turned ON with the help of a relay module which shows the concept of controlling high-power devices, i.e., AC bulbs. Relay is used as an electric switch actuated by the output signal from the microcontroller. In order to obtain smooth operation of light illumination, time delay is incorporated in between each of the lights' ON operations. Once the user reaches the other end, and the second sensor senses this, the controller turns OFF all the lights automatically. It is important to note that this approach allows the lights to stay ON only when needed, making it an efficient design for a smart house application. In the given project, the concepts of digital sensing, conditionals, sequencing, and relay interfacing are involved.

Moreover, it becomes clear that the usage of INPUT_PULLUP will provide correct operation since there will be no problem with floating input signals.

Circuit Diagram:

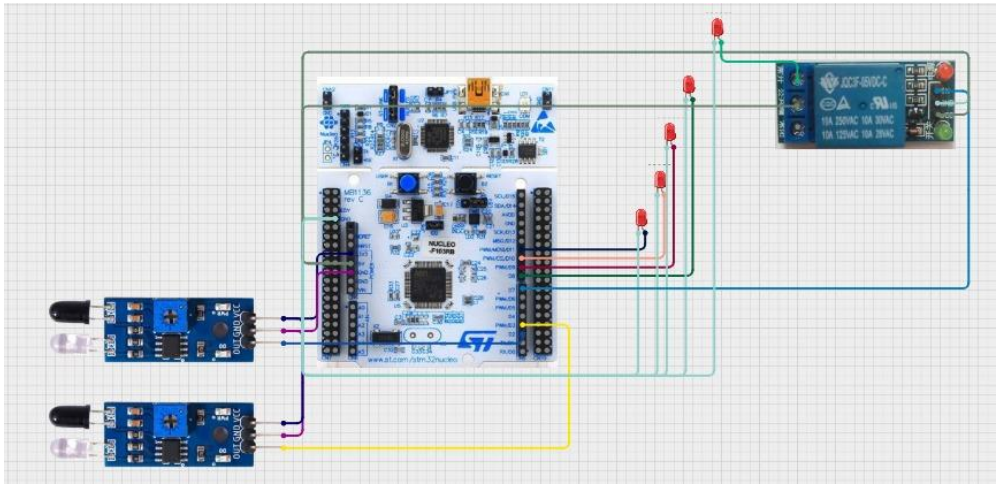


Fig1: Circuit diagram of the System

Simulation Results:



Fig2:Simulation results

Code:

```
// -------- PIN DEFINITIONS --------
#define S1  D2
#define S2  D3

#define LED1 D8
#define LED2 D9
#define LED3 D10
#define LED4 D11

#define RELAY D7   // LED5 via relay

// -------- SETUP --------
void setup()
{
  pinMode(S1, INPUT_PULLUP);
  pinMode(S2, INPUT_PULLUP);

  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);
  pinMode(LED3, OUTPUT);
  pinMode(LED4, OUTPUT);

  pinMode(RELAY, OUTPUT);

  //  RELAY OFF (ACTIVE HIGH → LOW = OFF)
  digitalWrite(RELAY, LOW);

  // LEDs OFF
  digitalWrite(LED1, LOW);
  digitalWrite(LED2, LOW);
  digitalWrite(LED3, LOW);
  digitalWrite(LED4, LOW);

  Serial.begin(9600);
}

// -------- FUNCTIONS --------

// UP → 1 → 2 → 3 → 4 → 5
void turnOnUp()
{
  digitalWrite(LED1, HIGH); delay(500);
  digitalWrite(LED2, HIGH); delay(500);
  digitalWrite(LED3, HIGH); delay(500);
  digitalWrite(LED4, HIGH); delay(500);
```

```
  digitalWrite(RELAY, HIGH); // LED5 ON
}

// DOWN → 5 → 4 → 3 → 2 → 1
void turnOnDown()
{
  digitalWrite(RELAY, HIGH); delay(500); // LED5 ON

  digitalWrite(LED4, HIGH); delay(500);
  digitalWrite(LED3, HIGH); delay(500);
  digitalWrite(LED2, HIGH); delay(500);
  digitalWrite(LED1, HIGH);
}

// OFF ALL
void turnOffAll()
{
  digitalWrite(LED1, LOW);
  digitalWrite(LED2, LOW);
  digitalWrite(LED3, LOW);
  digitalWrite(LED4, LOW);

  digitalWrite(RELAY, LOW); // LED5 OFF
}

// -------- LOOP --------
void loop()
{
  // -------- UP --------
  if (digitalRead(S1) == LOW)
  {
    Serial.println("UP");

    turnOnUp();

    while (digitalRead(S2) != LOW)
    {
      delay(10);
    }

    turnOffAll();
    delay(500);
  }

  // -------- DOWN --------
  if (digitalRead(S2) == LOW)
  {
    Serial.println("DOWN");
```

```
    turnOnDown();

    while (digitalRead(S1) != LOW)
    {
      delay(10);
    }

    turnOffAll();
    delay(500);
  }
}
```

Practical Results:

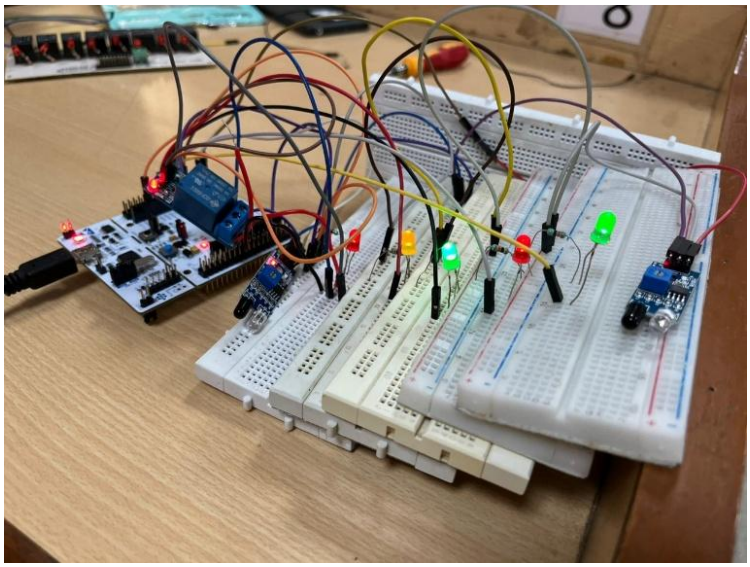


Fig3: Hardware Connections for the system

Conclusion:

The successful design and implementation of a smart staircase light automation system that is based on ARM Cortex-M4 with the help of double IR motion sensors, LEDs, and relay-operated lighting arrangement was achieved in this project. The system has been successfully capable of detecting human motion in both directions and automatically controlling the stairs light accordingly. This project has provided us with hands-on experience in designing an embedded system with sensor interfacing, digital I/O operations, and relay functioning with the help of the STM32F446RE microcontroller development board. It will not only minimize the wastage of energy but will also be user-friendly in nature.